01 / POWER DISTRIBUTION

Supply: external regulated 24 V DC. Nominal 1 A fused control branch.

Blue = signal; red = +24 V; black = 0 V. 18 AWG simulated cuts; no current-rating claim.

J0:+	W130 / 18 AWG / red / .25 m	F1:1
F1:2	W131 / 18 AWG / red / .25 m	D24:IN
J0:-	W132 / 18 AWG / black / .25 m	D0:IN
D24:10	W133 / 18 AWG / red / .5 m	PLC1:+24
D0:7	W134 / 18 AWG / black / .5 m	PLC1:0V

D24 and D0 are internally common distribution blocks; each numbered port takes one wire.

X1-X15 are two-sided feedthrough terminals: F (field) is internally connected to P (PLC).

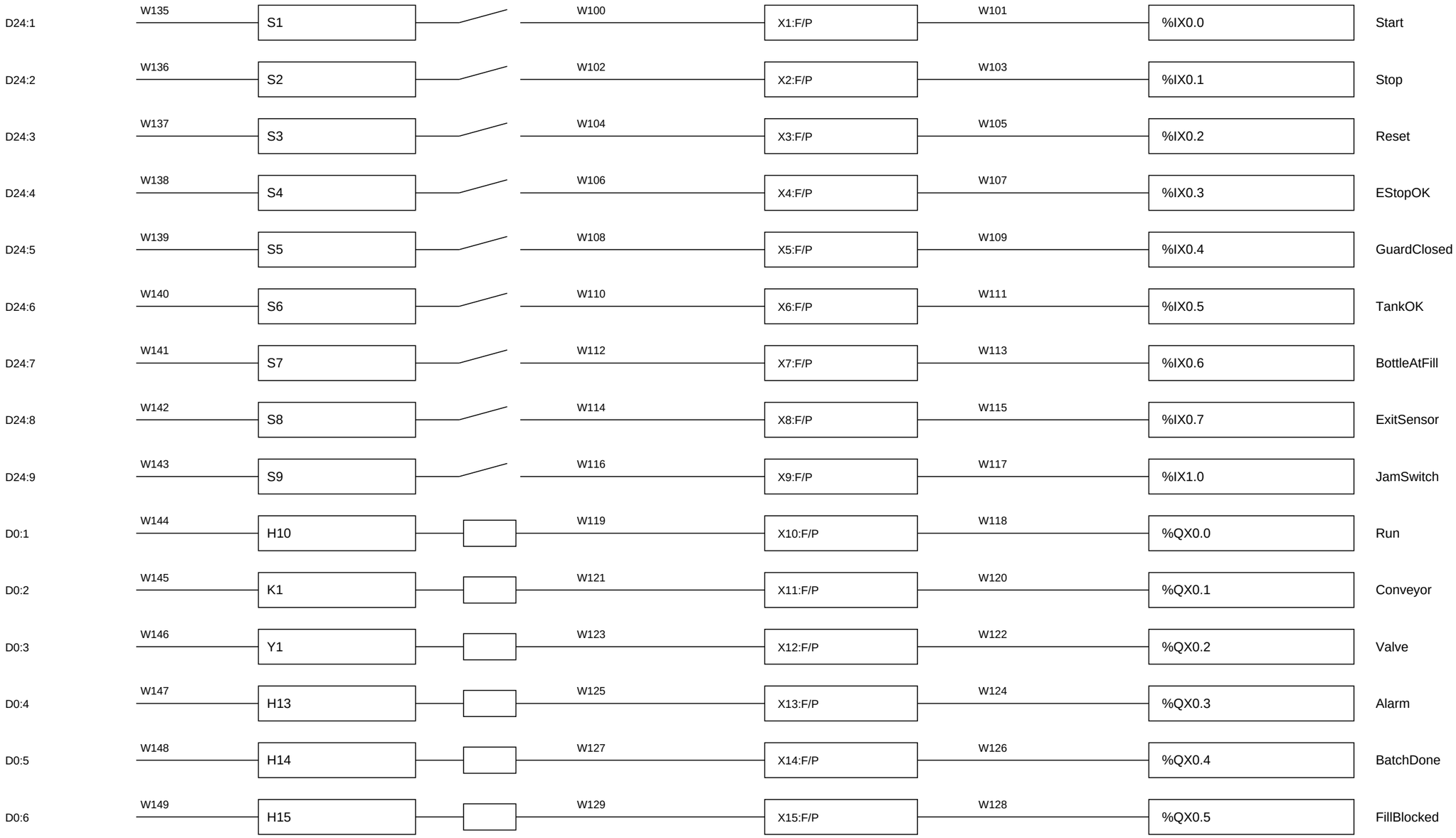
K1 is a contactor coil command only. Motor power and overload protection are outside this drawing.

EStopOK and GuardClosed are simulated healthy contacts, not a safety-rated stop chain.

For each branch, use its wire IDs on the I/O sheet and full endpoint schedule.

02 / 24 V DC INPUT AND OUTPUT WIRING

Each PLC address maps to one wired pin. Signal names match plc/line.xml.



Inputs: +24 V -> contact -> X:F/P -> PLC. Outputs: PLC -> X:P/F -> coil/lamp -> 0 V.

03 / TERMINAL AND WIRE SCHEDULE

PIN	WIRE	OTHER END	COLOUR	PIN	WIRE	OTHER END	COLOUR
S1:2	100	X1:F	blue	PLC1:%QX0.1	120	X11:P	blue
X1:F	100	S1:2	blue	X11:P	120	PLC1:%QX0.1	blue
X1:P	101	PLC1:%IX0.0	blue	X11:F	121	K1:1	blue
PLC1:%IX0.0	101	X1:P	blue	K1:1	121	X11:F	blue
S2:2	102	X2:F	blue	PLC1:%QX0.2	122	X12:P	blue
X2:F	102	S2:2	blue	X12:P	122	PLC1:%QX0.2	blue
X2:P	103	PLC1:%IX0.1	blue	X12:F	123	Y1:1	blue
PLC1:%IX0.1	103	X2:P	blue	Y1:1	123	X12:F	blue
S3:2	104	X3:F	blue	PLC1:%QX0.3	124	X13:P	blue
X3:F	104	S3:2	blue	X13:P	124	PLC1:%QX0.3	blue
X3:P	105	PLC1:%IX0.2	blue	X13:F	125	H13:1	blue
PLC1:%IX0.2	105	X3:P	blue	H13:1	125	X13:F	blue
S4:2	106	X4:F	blue	PLC1:%QX0.4	126	X14:P	blue
X4:F	106	S4:2	blue	X14:P	126	PLC1:%QX0.4	blue
X4:P	107	PLC1:%IX0.3	blue	X14:F	127	H14:1	blue
PLC1:%IX0.3	107	X4:P	blue	H14:1	127	X14:F	blue
S5:2	108	X5:F	blue	PLC1:%QX0.5	128	X15:P	blue
X5:F	108	S5:2	blue	X15:P	128	PLC1:%QX0.5	blue
X5:P	109	PLC1:%IX0.4	blue	X15:F	129	H15:1	blue
PLC1:%IX0.4	109	X5:P	blue	H15:1	129	X15:F	blue
S6:2	110	X6:F	blue	J0:+	130	F1:1	red
X6:F	110	S6:2	blue	F1:1	130	J0:+	red
X6:P	111	PLC1:%IX0.5	blue	F1:2	131	D24:IN	red
PLC1:%IX0.5	111	X6:P	blue	D24:IN	131	F1:2	red
S7:2	112	X7:F	blue	J0:-	132	D0:IN	black
X7:F	112	S7:2	blue	D0:IN	132	J0:-	black
X7:P	113	PLC1:%IX0.6	blue	D24:10	133	PLC1:+24	red
PLC1:%IX0.6	113	X7:P	blue	PLC1:+24	133	D24:10	red
S8:2	114	X8:F	blue	D0:7	134	PLC1:0V	black
X8:F	114	S8:2	blue	PLC1:0V	134	D0:7	black
X8:P	115	PLC1:%IX0.7	blue	D24:1	135	S1:1	red
PLC1:%IX0.7	115	X8:P	blue	S1:1	135	D24:1	red
S9:2	116	X9:F	blue	D24:2	136	S2:1	red
X9:F	116	S9:2	blue	S2:1	136	D24:2	red
X9:P	117	PLC1:%IX1.0	blue	D24:3	137	S3:1	red
PLC1:%IX1.0	117	X9:P	blue	S3:1	137	D24:3	red
PLC1:%QX0.0	118	X10:P	blue	D24:4	138	S4:1	red
X10:P	118	PLC1:%QX0.0	blue	S4:1	138	D24:4	red
X10:F	119	H10:1	blue	D24:5	139	S5:1	red
H10:1	119	X10:F	blue	S5:1	139	D24:5	red

CSV and Excel schedules include PLC signal names, addresses and wire gauge.

03 / TERMINAL AND WIRE SCHEDULE

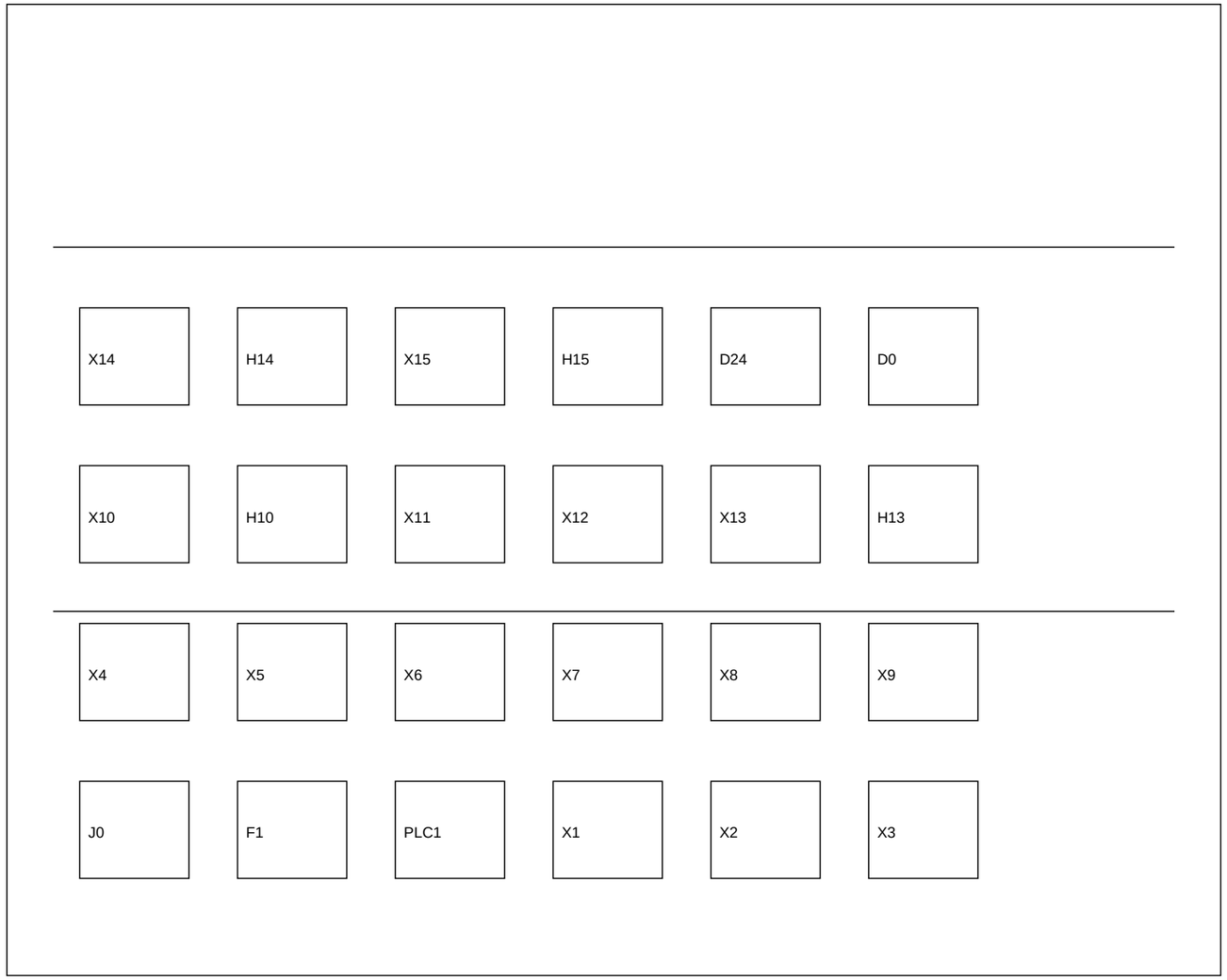
PIN	WIRE	OTHER END	COLOUR	PIN	WIRE	OTHER END	COLOUR
D24:6	140	S6:1	red				
S6:1	140	D24:6	red				
D24:7	141	S7:1	red				
S7:1	141	D24:7	red				
D24:8	142	S8:1	red				
S8:1	142	D24:8	red				
D24:9	143	S9:1	red				
S9:1	143	D24:9	red				
H10:2	144	D0:1	black				
D0:1	144	H10:2	black				
K1:2	145	D0:2	black				
D0:2	145	K1:2	black				
Y1:2	146	D0:3	black				
D0:3	146	Y1:2	black				
H13:2	147	D0:4	black				
D0:4	147	H13:2	black				
H14:2	148	D0:5	black				
D0:5	148	H14:2	black				
H15:2	149	D0:6	black				
D0:6	149	H15:2	black				

CSV and Excel schedules include PLC signal names, addresses and wire gauge.

04 / CONTROL PANEL LAYOUT

500 x 400 mm backplate study. Field sensors, contactor and valve are external.

Coordinates and footprints come from devices.csv; dimensions are illustrative millimetres.



PANEL REFERENCES

J0: INLET-24

F1: FUSE-1A

PLC1: PLC-24

X1: TB-2

X2: TB-2

X3: TB-2

X4: TB-2

X5: TB-2

X6: TB-2

X7: TB-2

X8: TB-2

X9: TB-2

X10: TB-2

H10: LED-24

X11: TB-2

X12: TB-2

X13: TB-2

H13: LED-24

X14: TB-2

H14: LED-24

X15: TB-2

H15: LED-24

D24: DIST-24

D0: DIST-0

Reserve wiring ducts and clearances before any physical construction; no enclosure or heat study included.

05 / CONSOLIDATED BILL OF MATERIALS

Device quantities are consolidated by part. Wire totals are cut lengths, without spare/waste allowance.

PART	DESCRIPTION	QTY	UNIT
CONTACT-24	24 V simulated dry contact	9	each
DIST-0	Internally common 0 V distribution	1	each
DIST-24	Internally common +24 V distribution	1	each
FUSE-1A	1 A control fuse	1	each
INLET-24	24 V DC supply inlet	1	each
LED-24	24 V panel indicator	4	each
PLC-24	Simulated 24 V PLC with 9 inputs and 6 outputs	1	each
RELAY-24	24 V conveyor contactor coil	1	each
TB-2	Two-sided feedthrough terminal	15	each
VALVE-24	24 V filling-valve solenoid	1	each
WIRE:18 AWG:black	18 AWG black hookup wire	4.35	m
WIRE:18 AWG:blue	18 AWG blue hookup wire	12	m
WIRE:18 AWG:red	18 AWG red hookup wire	6.4	m

Generic study parts are not manufacturer selections; PLC-24 represents the simulated controller, not a selected commercial model.